DHAR

Volume II — Engineering Roadmap

STAGE-EPSILON

“Deploying Dhar — trustworthy enough for infrastructure that can't fail quietly.”

Document	Stage-epsilon — Deploying Dhar
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Part of	The Canon of Dhar, Volume II
Depends on	Stage-delta
Feeds into	Tempering Dhar (Volume III)

1. Mandate

Make Dhar trustworthy enough for production use, including the critical-infrastructure and defence use cases Spec0 §5 names. This is the stage Tempering Dhar (Volume III) leans on most heavily, and the last of the five engineering stages before work moves to the ecosystem and program-management volumes.

2. Scope

- **Cross-compilation** across all five backends named in Spec0 §4: x86-64, ARM64, RISC-V, WASI, Windows.
- **Debug information** generation for source-level debugging.
- **Profiling hooks** built into the runtime.
- **Static analysis** tooling beyond the compiler's own type checking.
- **Fuzzing infrastructure**, covering both the compiler's own front end and compiled Dhar programs.
- **Reproducible builds**: bit-for-bit identical output from the same source on different machines.
- **SBOM generation** for every release, built directly from Kun's own dependency-resolution records (Stage-delta §3.7) rather than reconstructed separately after the fact.
- **Signing and supply-chain tooling**, with a documented verification path.

3. Required Architecture

Every backend added at this stage **must** consume the same Universal IR Stage-beta and Stage-gamma already established (Spec0 §4). A backend that requires changes to the IR itself, the parser, or the type system is a violation of the independence commitment in Spec0 §2.2 — the whole point of the IR boundary is that a new target is only ever a new backend module.

4. Acceptance Criteria

1. A single, unmodified source tree cross-compiles to working binaries on all five backends.
2. Two independent builds of the same source, on two different machines, are bit-for-bit identical.
3. Every tagged release ships with a generated SBOM.
4. Every release binary is signed, with a documented, independently verifiable path.
5. Fuzzing infrastructure runs continuously against the compiler's own front end (Stage-beta/gamma) as well as against compiled Dhar programs.

5. Handoff

Once Stage-epsilon's acceptance criteria are met, Dhar is specified as production-ready. Ongoing work moves to the ecosystem volumes (Volume III) and program management (Volume IV, The Pact of Developing Dhar) rather than further core-engineering stages — the five-stage roadmap that began at Stage-alpha ends here.

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